Liquid intrusion and evaporation in low-pressure cavities: numerical investigation with STAR-CCM+

Einströmung und Verdampfung einer Flüssigkeit unter Niedrigdruckbedingungen: numerische Simulationen mit STAR-CCM+

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Masterarbeit

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1 Numerical Methods

This chapter contains an overview of the governing equations used in Star-CCM+ for the modelling of multiphase flows with a focus on eulerian multiphase (EMP) flow and the volume of fluid (VOF) method. Turbulence and the Shear Stress Transport (SST) k-ω model is briefly discussed. The physical processes of evaporation and vaporisation are described. The ability to model multiple flow regimes, capture interfaces, and the phase interaction closures used in Star-CCM+ are discussed. The chapter concludes with a description of the numerical methods used in Star-CCM+ along with the apparent limitations [better word] of the commercial software in being able to accurately model the problems under consideration.

1.1 Governing Equations

[define region as in what is V and what is delV and stuff] Multiphase flow is modelled in Star-CCM+ using various frameworks. The two-fluid - eulerian multiphase - framework solves a set of six equations to model two-phase flow. A set of transport equations for mass, momentum, and energy are solved for all the phases, which are assumed to be interpenetrating continua that share a common pressure field. The local instantaneous conservation equations are assumed to be euler averaged and contain closures based on interfacial terms that define the influences of the phases upon each other.

Although used primarily for modelling dispersed flows, the EMP framework is capable of resolving both dispersed and separated flow regimes.

The implemented volume of fluid method in Star-CCM+ assumes sufficient mesh resolution in order to resolve the interface and model separated flows. The interface capturing method solves the volume fraction transport equation using the High-Resolution Interface Capturing (HRIC) scheme to maintain a sharp interface between the phases.

Based on the volume fraction of the phase i, $\alpha_i = V_i/V$, where in each cell $\sum_i \alpha_i = 1$. The material properties in each cell and other flow variables are assumed to be a mixture of the properties of the phases as follows:

$$\text{density, } \rho = \textstyle\sum_i \alpha_i \rho_i, \tag{1.1a}$$

$$\text{viscosity, } \mu = \textstyle\sum_i \alpha_i \mu_i, \tag{1.1b}$$

$$\text{velocity, } \underline{\mathbf{v}} = \textstyle\sum_i \alpha_i \rho_i \underline{\mathbf{v}}_i / \rho, \tag{1.1c}$$

$$\text{specific heat, } C_p = \textstyle\sum_i \alpha_i C_{p,i} \rho_i / \rho, \tag{1.1d}$$

$$\text{thermal conductivity, } k = \textstyle\sum_i \alpha_i k_i, \tag{1.1e}$$

$$\text{enthalpy, } \mathrm{H} = \textstyle\sum_i \alpha_i \rho_i \mathrm{H}_i / \rho + |\underline{\mathbf{v}}|^2/2. \tag{1.1f}$$

where ρ_i is the density, μ_i is the dynamic viscosity, $\underline{\mathbf{v}}_i$ is the velocity, $C_{p,i}$ is the specific heat, k is the thermal conductivity, and H_i is the enthalpy of the phase i.

The diffusion velocity of phase i, $\underline{\mathbf{v}}_{d,i}$, is defined as the difference between the velocity of the phase and the mixture such that

$$\underline{\mathbf{v}}_{d,i} = \underline{\mathbf{v}}_i - \underline{\mathbf{v}}.$$

A set of volume fraction transport and mass, momentum, and energy conservation equations are solved for each phase. In cases of two-phase flow, Star-CCM+ solves the volume fraction transport equation only for the first phase.

The governing equations used in the implementation of both the EMP and VOF frameworks in Star-CCM+ are summarised below.

1.1.1 Eulerian Multiphase Flow

The **mass conservation** equation solved for the phase i is

$$\frac{\partial}{\partial t}\int_V \alpha_i\rho_i \,\mathrm{d}V + \int_{\partial V} \alpha_i\rho_i\underline{\mathbf{v}}_i\cdot \mathrm{d}\underline{\mathbf{a}} = \int_V \sum_{j\neq i}(m_{ij}-m_{ji})\,\mathrm{d}V + \int_V \mathrm{S}_i^{\alpha}\,\mathrm{d}V \tag{1.2}$$

[CHECK]where for the phase i

- $\underline{\mathbf{a}}$ is the surface area vector,
- α_i is the volume fraction that fulfills the requirement $\sum_i^n \alpha_i = 1$,
- ρ_i is the density,
- $\underline{\mathbf{v}}_i$ is the velocity,
- m_{ij}, m_{ji} is the interphase mass transfer per unit volume from phase i to j and j to i respectively,
- S_i^{α} is the user defined mass source term.

The **momentum conservation** equation solved for the phase i is

$$\begin{aligned}\frac{\partial}{\partial t}\int_V \alpha_i\rho_i\underline{\mathbf{v}}_i\,\mathrm{d}V + \int_{\partial V}\alpha_i\rho_i\underline{\mathbf{v}}_i\otimes\underline{\mathbf{v}}_i\cdot\mathrm{d}\underline{\mathbf{a}} = &-\int_V \alpha_i\nabla p\,\mathrm{d}V + \int_V \alpha_i\rho_i\underline{\mathbf{g}}\,\mathrm{d}V + \int_{\partial V}\left[\alpha_i\left(\underline{\underline{\mathbf{T}}}_i + \underline{\underline{\mathbf{T}}}_i^t\right)\right]\cdot\mathrm{d}\underline{\mathbf{a}} + \int_V \underline{\mathbf{M}}_i\,\mathrm{d}V\\ &+\int_V (\underline{\mathbf{F}}_{\mathrm{int}})_i\,\mathrm{d}V + \int_V \underline{\mathbf{S}}_i^{\alpha}\,\mathrm{d}V + \int_V\sum_{j\neq i}(m_{ij}\underline{\mathbf{v}}_j - m_{ji}\underline{\mathbf{v}}_i)\,\mathrm{d}V\end{aligned} \tag{1.3}$$

where for the phase i

- p is the pressure,
- $\underline{\mathbf{g}}$ is the gravitational force vector,
- $\underline{\underline{\mathbf{T}}}_i, \underline{\underline{\mathbf{T}}}_i^t$ are the molecular and turbulent stresses respectively,
- $\underline{\mathbf{M}}_i$ is the interphase momentum transfer term per unit volume that satisfies $\sum_{i=1}^n \underline{\mathbf{M}}_i = 0$,
- $(\underline{\mathbf{F}}_{\mathrm{int}})_i$ is the internal force vector that consists of either the solid pressure force between particles or user defined potential force,
- $\underline{\mathbf{S}}_i^{\alpha}$ is the user defined momentum source term.

The **energy conservation** equation solved for the phase i is

$$\begin{aligned}\frac{\partial}{\partial t}\int_V \alpha_i\rho_i\mathrm{E}_i\,\mathrm{d}V + \int_{\partial V}\alpha_i\rho_i\mathrm{H}_i\underline{\mathbf{v}}_i\cdot\mathrm{d}\underline{\mathbf{a}} = &\int_{\partial V}\alpha_i(k_{\mathrm{eff}})_i\nabla T_i\cdot\mathrm{d}\underline{\mathbf{a}} + \int_{\partial V}\underline{\underline{\mathbf{T}}}_i\underline{\mathbf{v}}_i\cdot\mathrm{d}\underline{\mathbf{a}} + \int_V \underline{\mathbf{f}}_i\cdot\underline{\mathbf{v}}_i\,\mathrm{d}V + \int_V \mathrm{S}_{u,i}\,\mathrm{d}V\\ &+\int_V\sum_{j\neq i}Q_{ij}\,\mathrm{d}V + \int_V\sum_{ij}Q_i^{(ij)}\,\mathrm{d}V + \int_V\sum_{j\neq i}(m_{ij}-m_{ji})h_i(T_{ij})\,\mathrm{d}V\end{aligned} \tag{1.4}$$

where for the phase i,

- E_i is the total energy,
- H_i is the total enthalpy,
- $\underline{\underline{\mathbf{T}}}_i$ is the viscous stress tensor,
- T_i is the temperature,
- $\underline{\mathbf{f}}_i$ is the body force vector,
- $(k_{\mathrm{eff}})_i$ is the effective thermal conductivity,

- $\mathrm{S}_{u,i}$ is the user defined energy source term,
- Q_{ij} is the heat transfer from phase i to phase j due to temperature difference between the phases,
- $Q_i^{(ij)}$ is the heat transfer between the phase i and the interface which accounts for phenomenon such as evaporation and boiling,
- $h_i(T_{ij})$ is the enthalpy of the phase i evaluated at the temperature T_{ij} of the interface.

The effective thermal conductivity of the phase i is defined using the thermal conductivity, k_i, the turbulent viscosity, $\mu_{t,i}$, the specific heat, $C_{p,i}$, and the turbulent thermal diffusion Prandtl number, $\sigma_{t,i}$ as

$$(k_{\text{eff}})_i = k_i + \frac{\mu_{t,i} C_{p,i}}{\sigma_{t,i}}$$

.

The total enthalpy of the phase i, H_i, is defined using the specific enthalpy h_i and the kinetic energy as

$$\mathrm{H}_i = h_i + \frac{1}{2}\underline{\mathbf{v}}_i \cdot \underline{\mathbf{v}}_i,$$

where the specific enthalpy is further defined using the specific heat of the phase and the reference temperatures and heat of formation such that

$$h_i = h_i^{\text{REF}} + \int_{T_i^{\text{REF}}}^{T_i} C_{p,i}(T')\,\mathrm{d}T'.$$

The total energy of the phase i is then related to the total enthalpy as $\mathrm{E}_i = \mathrm{H}_i - p/\rho_i$.

The interphase energy terms are related as follows:

- for cases where heat is exchanged only through thermal diffusion, the interphase heat transfer rates satisfy $Q_{ij} = -Q_{ji}$
- for cases where heat and mass are transferred only through phase change, the interphase heat and mass transfer rates satisfy $Q_i^{(ij)} + Q_j^{(ij)} + \left(m_{ij} - m_{ji}\right)\left(h_j(T_{ij}) - h_i(T_{ij})\right) = 0$.

1.1.2 Volume of Fluid Method

The distribution of the phase i is driven by the volume fraction transport.

$$\frac{\partial}{\partial t}\int_V \alpha_i\,\mathrm{d}V + \int_{\partial V} \alpha_i \underline{\mathbf{v}}_i \cdot \mathrm{d}\underline{\mathbf{a}} = \int_V \left(\mathrm{S}_i^\alpha - \frac{\alpha_i}{\rho_i}\frac{D\rho_i}{Dt}\right)\mathrm{d}V - \int_V \frac{1}{\rho_i}\nabla\left(\alpha_i \rho_i \underline{\mathbf{v}}_{d,i}\right)\mathrm{d}V \tag{1.5}$$

where $\underline{\mathbf{a}}$ is the surface area vector, $\underline{\mathbf{v}}$ is the mixture velocity, $\underline{\mathbf{v}}_{d,i}$ is the slip velocity, and S_i^α is the user defined phase source term for the phase i.

The **mass conservation** equation solved is

$$\frac{\partial}{\partial t}\int_V \rho\,\mathrm{d}V + \int_{\partial V} \rho \underline{\mathbf{v}} \cdot \mathrm{d}\underline{\mathbf{a}} = \int_V \sum_i \mathrm{S}_i^\alpha \rho_i\,\mathrm{d}V \tag{1.6}$$

The **momentum conservation** equation solved is

$$\begin{aligned}\frac{\partial}{\partial t}\int_V \rho\underline{\mathbf{v}}\,\mathrm{d}V + \int_{\partial V} \rho\underline{\mathbf{v}} \otimes \underline{\mathbf{v}} \cdot \mathrm{d}\underline{\mathbf{a}} = -\int_V \nabla p\,\mathrm{d}V + \int_V \rho\underline{\mathbf{g}}\,\mathrm{d}V + \int_{\partial V}\left[\left(\underline{\underline{\mathbf{T}}} + \underline{\underline{\mathbf{T}}}^T\right)\right]\cdot \mathrm{d}\underline{\mathbf{a}} + \int_V \underline{\mathbf{f}}_b\,\mathrm{d}V \\ - \sum_i \int_{\partial V} \alpha_i \rho_i \underline{\mathbf{v}}_{d,i} \otimes \underline{\mathbf{v}}_{d,i} \cdot \mathrm{d}\underline{\mathbf{a}} + \int_V \underline{\mathbf{S}}_i^\alpha\,\mathrm{d}V\end{aligned} \tag{1.7}$$

where

- p is the pressure,
- $\underline{\mathbf{g}}$ is the gravitational force vector,
- $\underline{\underline{\mathbf{T}}}$ is the viscous stress tensor,
- $\underline{\mathbf{f}}_b$ is the body force vector,
- $\underline{\mathbf{S}}_i^\alpha$ is the user defined phase momentum source term.

The **energy conservation** equation solved is

$$\frac{\partial}{\partial t}\int_V \rho \mathrm{E}\,\mathrm{d}V + \int_{\partial V}\left[\rho \mathrm{H}\underline{\mathbf{v}} + \sum_i \alpha_i \rho_i \mathrm{H}_i \underline{\mathbf{v}}_{d,i}\right]\cdot \mathrm{d}\underline{\mathbf{a}} = -\int_{\partial V} \underline{\dot{\mathbf{q}}''}\cdot \mathrm{d}\underline{\mathbf{a}} + \int_{\partial V} \underline{\underline{\mathbf{T}}}\cdot \underline{\mathbf{v}}\,\mathrm{d}\underline{\mathbf{a}} + \int_V \underline{\mathbf{f}}_b \cdot \underline{\mathbf{v}}\,\mathrm{d}V + \int_V \mathrm{S}_\mathrm{E}\,\mathrm{d}V \tag{1.8}$$

where

- E is the total energy,
- H is the total enthalpy,
- $\underline{\dot{\mathbf{q}}''}$ is the heat flux vector,
- $\underline{\underline{\mathbf{T}}}$ is the viscous stress tensor,
- S_E is the user defined energy source term.

1.2 Modelling Turbulence

Flows are generally turbulent. Turbulence is chracterised by irregular fluid motion dominated by eddies and vortices across a wide range of spatial and temporal scales. The largest eddies extract kinetic energy from the mean flow, which is then, through a cascade, transferred to the eddies at the smallest scale where the energy is dissipated due to viscosity, raising the internal energy of the fluid [1, 2].

Although turbulence can be modelled in various ways to various levels of resolutions, turbulence in this study has been modelled using the Shear Stress Transport (SST) k-w model. The SST k-w model is a two-equation eddy-viscosity turbulence model based on the Reynolds-Averaged Navier-Stokes (RANS) equations. The model combines the advantages of the k-ω model near walls—which accurately resolves boundary layer behaviour—and the k-ε model in the free stream, enhancing robustness and accuracy in separated flows [3]. The SST approach uses a blending function to smoothly transition between these models, which is especially effective in flows with strong adverse pressure gradients and separation, such as multiphase cavity flows [Menter1994].

RANS models result from decomposing flow variables into mean and fluctuating components,$\phi = \bar{\phi} + \phi'$, applying Reynolds averaging, which filters out turbulent fluctuations below the averaging scale [4, 5]. This operation yields additional Reynolds stress terms representing momentum transport by turbulence, which require closure. In order to draw closure relations, eddy-viscosity models of turbulence assume that:

- the turbulent scalar flux vector aligns with the mean scalar gradient.
- the Reynolds stress tensor is proportional to the mean velocity strain rate through an eddy viscosity.

While these assumptions simplify modelling, they tend to be, more often than not, invalid as they do not fully capture turbulence physics, especiialy in anisotropic or non-equilibrium flow [6]. This limitation can be significant in regions with complex strain rates or strong secondary flows.

The implementation of the model in Star-CCM+ can be further controlled to adjust the realisability. Realisability is the postive semi-definiteness of the reynolds stress tensor. The specification of turbulence is done using the turbulence intensity and the length scale. The inherent anisotropy in turbulence is accounted for with quadratic and cubic constitutive options in the modelling of reynolds stresses. The parameters in the model are left on their default values as they are empirically modelled.

1.3 Evaporation

The evaporation process is distinguished between occuring at saturation temperatures or not. Evaporation at saturation temperatures, known as vaporisation, is a much faster process in comparison to evaporation at normal [7]. In case of vaporisation along a heated wall, most CFD models of vaporisation treat the energy transfer by partitioning the heat flux at the wall into the latent heat of the bubble vaporisation process and the sensible heat transferred to the liquid through conduction, convection, and quencing processes. This is determined by the area fractions covered by the liquid and vapour. Correlations between the heat flux and wall superheat [what is superheat] have been made using empirical results or based on modelling of sub-processes.

The sub-processes that are primarily taken into account in such determinations are:

- number density of nucleation sites
- diameter of bubble at time of detachment from the surface
- frequency of bubble detachment

Nucleation affects the number and distribution of bubbles in the fluid. Walls, particularly so in case they're heated, are more suspectible to be nucleation sites for bubble growth as the activation energy for bubble growth is much lower. This nucleation can be modelled through the domain as homogenous or heterogenous - where nucleation sites are modelled using a presumed PDF function or be determined by empirical correlations. [8]

After a stable bubble nucleus has been formed, the growth of the bubble becomes pertinent in the evaluation of the diameter at departure and the interphase energy and mass transfer. In the case of an ideal spherical bubble undergoing expansion in a uniformly superheated liquid, the rate of growth is determined by the surface tension, liquid inertia, and the differnce in pressures inside and outside the bubble. Surface tension dominates and restricts the initial bubble growth. This initial rapid expansion of the bubble results in the formation of a thin liquid layer separating the vapour from the wall surface. This microlayer can be a significant contributor to the bubble growth and heat transfer process. As the radius increases and the liquid layer around the bubble cools due to evaporation, a decrease in vapour pressure in seen. The dynamics are controlled by thermal diffusion and inertia. Inertial effects, however, become less relevant as the bubble growth rate decreases and the bubble radius grows. The interphase energy exchange determines further bubble growth. [9]

1.3.1 Effects of low pressure and gravity conditions

The absence of significant gravitaional forces alters the dynamics of vaporisation - affecting natural convection, buoyancy driven phenomenon, and bubble dynamics. Under terrestrial gravity, convection currents develop under thermal gradients, thus bringing cooler liquid to the heated surface as the vapour bubble rises due to buyoancy. This density driven natural convection is affected in low gravity conditions, maintaining the thermal gradient. Bulk fluid movement is affected and heat and mass transfer processes shift from being convection dominated to being diffusion dominated. [10, 11]

Under low gravity conditions, vapour bubbles tend to remain near the heat source, growing signifcantly larger and coalescing with smaller bubbles. Prolonged bubble presence on the surface results in drying out of the surface and unsteady rise in surface temperatures.[12] [Mention some numbers] In conditions where the dominance of surface tension forces becomes more pronounced, the surfae tension gradients can induce fluid circulation. This is known as Marangoni convection and is observed to affect bubble growth and heat transfer. [13]

Reduced system pressure impacts the overall intensity of heat transfer. The onset of nucleate boiling also requires higher superheat values in low pressure conditions. Decrease in pressure leads to a lower number of active nucleation sites and increases microlayer thickness found at the bubble-wall interface. The local heat flux density from microlayer evaporation is impacted and the microlayer evaporation area and depletion time is increased. The system is also susceptible to the formation of a continuous vapour film at the surface due to coalescing bubbles unable to escape the surface, reducing the overall heat transfer. [14]

The affected heat transfer and bubble growth phenomenon further result in lower critical heat fluxes and heat transfer efficiency. [15, 16]

1.3.2 Vaporisation in a superheated liquid

[————UNDER CONSTRUCTION————————] Flash evaporation is the intense vaporisation triggered through depressurisation of superheated fluids.

1.4 Multiple Flow Regimes and Interface Capturing

Star-CCM+ in its implementation of the eulerian-multiphase framework utilises a dynamic flow regime topology based on the local volume fractions of the interacting phases. Based on the increasing volume fraction of the secondary phase in a computational cell, the topology transitions through three principal regimes:

- dispersed in the primary phase
- having a large scale interface with the primary phase
- continuous while having the primary phase dispersed

For each cell, Star-CCM+ uses regime-specific closure models for the interfacial [OR RATHER INTERPHASE MOMENTUM AND ENERGY TRANSFER] forces. These models are blended using weighting schemes according to the calculated proportion of each regime in the cell. [17] [Gada et al. (2015): Large Scale Interface Model for Simulating Multiphase Flows (STAR-CCM+ Implementation)]

[EDIT]While there are multiple numerical methods to account for volume fraction transport, the High-Resolution Interface Capturing scheme is specifically implemented to handle sharp phase interfaces. Free-surface flows with sharp interfaces present a discontinuity in the solution domain, causing discretisation errors. This can lead to spurious or parasitic currents. [18] Although often negligible, these currents can be accommodated for through the incorporation of an artificial viscosity term.

At large scale interfaces, the near-interface flow often exhibits laminar characteristics with the development of boundary layers. However, in situations with significant velocity gradients, turbulent fluctuations can arise in the regions adjacent to the interface (especially when the interface is mobile or highly distorted). Recognizing that standard RANS turbulence models tend to overpredict turbulence intensity near phase boundaries, STAR-CCM+ allows the user to apply turbulence damping strategies (e.g., vorticity limiters) which suppress excessive eddy viscosity and restore physically consistent laminar sublayers adjacent to the interface.

1.5 Phase Interactions

[————UNDER CONSTRUCTION————————] Phase interactions in free-surface flows and flash evaporation dictate how mass, momentum and energy are exchanged between contiguous liquid and gaseous domains. For both separated (e.g., stratified) and dispersed (e.g., bubbly) flow regimes, major phase interactions include:

- Drag force: Governs momentum exchange due to the relative motion between phases; typically decomposed using empirical drag laws (e.g., Schiller-Naumann for bubbles/droplets).
- Lift force: Acts perpendicular to the relative flow, important in shear flows with small dispersed phases; affects lateral migration of bubbles/droplets—uses models like Tomiyama, Saffman, or others depending on flow characteristics.
- Virtual mass force (added mass) Represents resistance to acceleration caused by the acceleration of surrounding fluid (significant in cases with rapid acceleration or deceleration of bubbles).
- Wall lubrication force Repels dispersed phase from solid walls, preventing unrealistic accumulation at boundaries (important in vertical/horizontal pipe flows).

- Turbulent dispersion force Spreads the dispersed phase due to turbulence in the continuous phase, modeled typically as a diffusion term proportional to turbulent kinetic energy gradient. The Favre-Averaged Drag (FAD) model approximates turbulent dispersion as a function of liquid eddy viscosity and bubble slip. This mitigates excessive bubble clustering below the free surface. [19]
- Surface tension Maintains interface shape, forms basis for capillary effects and controls droplet/bubble formation and coalescence/breakup.
- Interphase mass transfer (evaporation/condensation) Represents phase change, modeled as interfacial flux determined by local thermodynamic properties (temperature, concentration, pressure—often using kinetic models for flash evaporation).
- Interphase energy transfer Encompasses latent heat and sensible heat exchange at the interface and throughout the phases (essential for accurate phase change modeling).
- Nucleation (wall and bulk)
- Interfacial area changes due to bubble growth, coalescence, breakup
- Local pressure and temperature gradients driving simultaneous mass and energy transfer

1.6 Numerical Methods

[————UNDER CONSTRUCTION————————]

- gradients
- solvers - implicit unsteady, segregated flow (SIMPLE)
- Relevant non-dim numbers? (reynolds, froude, jakobs, nusselt, bond, what not)

1.7 Workflow?

[————UNDER CONSTRUCTION————————]

2 Subproblem: Liquid Infiltration

[————UNDER CONSTRUCTION————–]

The first subproblem addresses water infiltrating the evaporation chamber during rocket ascent.

The flow system assembly comprises the hydraulic accumulator, the evaporation chamber, and the injection system subassembly that drives the liquid from the hydraulic accumulator into the evaporation chamber. The hydraulic accumulator and the evaporation chamber can be heated until lift-off. Current design considerations assume the temperature of the evaporation chamber and water in the accumulator to be 368.15 K and 353.15 K respectively. The ambient temperature at lift-off is assumed to be 280 K based on the remarks and considerations made in [OLD-SED].

Although insulation from the environment outside the rocket is assumed and the hydraulic accumulator will be sufficiently insulated, the evaporation chamber may cool down slightly during ascent until the start of the experiment at time $t = 140\,\mathrm{s}$ after lift-off. As of yet, no considerations have been made regarding the insulation of the evaporation chamber.

The pre-charge pressure in the accumulator and the initial pressure in the evaporation chamber are assumed to be 3.5 bar and 1 bar respectively. Water injection begins after the data storage signal is sent by the electronics module and stops shortly before the apogee is reached, when the absolute pressure in the evaporation chamber rises to 1.7 bar. Due to water remaining at the time of cutoff in the injection subassembly and also the accumulator, the total amount of water discharged to the chamber is assumed to be 150 mL.

No flow measurements are made in the injection assembly. Pressure and temperature are measured in the evaporation chamber along the nozzle at the midspan, the throat, and the exit.

2.1 Objective

The objective of the first task is to model the water injection into the evaporation chamber and extract quantities of interest to better estimate the initial conditions for the second task. The validity of the multiphase physics models provided by Star-CCM+ and the equations implemented therein is also evaluated against the one-dimensional Stefan problem and the two-dimensional Rayleigh-Taylor instability.

2.2 Validation Problems

2.2.1 Stefan Problem

Phase change processes are characterized by the movement of an interface separating two distinct phases, where one phase expands as another contracts due to thermal or non-thermal mechanisms [Koga, 2013]. The mathematical model governing such moving boundary problems is known as the Stefan system [Koga, 2013], and the analysis of such a system forms the basis of the Stefan problem.

In the classical Stefan problem, a fixed-temperature boundary propagates a melting or freezing front into the domain, with interface movement determined by the balance of latent and conducted heat. [20] With uniform initial temperature, constant boundary temperature and thermophysical properties in each phase, the semi-infinite 1D Stefan problem is the simplest, explicitly solvable problem of phase change. As the problem admits an analytical solution and reduces to the standard transient heat conduction problem when latent heat is neglected, it serves as a rigorous benchmark for validating phase-change modelling and the correct implementation of the energy equation. [20, 21, 22].

Assuming that the region in space, $\Omega \subset \mathbb{R}^2$, occupied by the material is the 2D semi-infinite rectangle $\Omega = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x \in [0, \infty), y \in [0, \Delta h]\}$, with the solid phase and the liquid phase occupying the regions Ω_s

and Ω_l respectively such that $\Omega = \Omega_s \cup \Omega_l$. At any given time, t, the two phases are separated by the interface, $X(t)$, that marks the change in phase of the material. The boundary of the domain Ω is denoted by $\partial\Omega$.

The problem space, $\mathcal{D} = \{[t_0, t_f] \times \Omega\}$, is the spatial and temporal domain occupied by the material. With the heat equation governing the temperature field in the liquid region, the Stefan condition at the interface, and the respective boundary conditions, the system of equations defining the one-dimensional Stefan problem can be written as,

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}, \text{ for } 0 < x < X(t), t_0 < t < t_f \tag{2.1a}$$

$$-k \frac{\partial T(X(t), t)}{\partial x}\Big|_{X^-(t)} = \rho L \frac{\partial X(t)}{\partial t}, \text{ for } t_0 < t < t_f \tag{2.1b}$$

$$T(x, 0) = T_m, \text{ on } \Omega \tag{2.1c}$$

$$T(0, t) = T_l, \text{ on } \partial\Omega(0, t) \text{ for } t_0 < t < t_f \tag{2.1d}$$

$$\nabla T \cdot \underline{\mathbf{n}} = 0, \text{ on } \partial\Omega - \partial\Omega(0, t) \tag{2.1e}$$

An analytical solution to the system of equations is given as

$$\hat{T}(x, t) = T_l - (T_l - T_m) \frac{\text{erf}\left(\frac{x}{2\sqrt{\alpha t}}\right)}{\text{erf}\left(\frac{X(t)}{2\sqrt{\alpha t}}\right)}, \tag{2.2}$$

$$\hat{X}(t) = 2\lambda\sqrt{\alpha t}, \tag{2.3}$$

where λ is the unique root of the monotonic function

$$f(\lambda) := \frac{C_p(T_m - T_l)}{L} - \frac{\lambda e^{-\lambda^2}}{\sqrt{\pi}\text{erf}(\lambda)}$$

The error function $\text{erf}(x)$ is defined as $\text{erf}(x) := 2/\sqrt{\pi} \int_0^x e^{-y^2}\, dy$. The physical parameters and computational methodology are summarized in Table 1

The simulations assume a laminar but unsteady flow. The computational domain of the simulation is chosen to be $[0, L] \times [0, 4\Delta h]$, where Δh is the cell width of the discretised domain. The domain is discretised using a trimmed cell mesher [CONFIRM] with uniform quadrilateral elements. The volume of fluid method with first order HRIC is used to reconstruct the interface. The convection and diffusion terms are discretised using the second-order upwind scheme and the least squares method was used to interpolate the values of the gradients from the cell centres onto the face centres. The transient term is discretised using the implicit Euler-method which is first-order accurate. All four boundaries are assigned a no-slip shear stress specification. The boundary at $x = 0$ is assigned a constant temperature, $T_l = 300$ K, while the other boundaries are assumed to be adiabatic. Since Star-CCM+ sets under- and over-relaxation factors conservatively, the values are left unchanged.

The physical parameters and the computational methods involved in the simulation setup are summarised in Table 1. The evolution of the temperature field for the simulation with cell size, $\Delta h = 5.0 \times 10^{-3}$ m is shown in Figure 1. The interface is observed to move towards the right with increasing simulation time. The temperature of the system can also be seen to decrease from the Dirichlet boundary at $x = 0$ towards the melting temperature, T_m, at the interface, after which it stays constant.

Simulation Setup	
Domain	2D rectangular domain
Mesh size, Δh	2.0×10^{-3} m
Time discretization	Implicit unsteady, 1st order Euler
Time step, Δt	0.5 s
Physical Properties	
Density, ρ	$1000.0\,\mathrm{kg{\cdot}m^{-3}}$
Dynamic viscosity, μ	8.8871×10^{-4} Pa·s
Specific heat, C_p	$4200.0\,\mathrm{J{\cdot}(kg{\cdot}K)^{-1}}$
Latent heat of fusion, L	$1.0 \times 10^{5}\,\mathrm{J{\cdot}kg^{-1}}$
Thermal conductivity, k_l	$2.33\,\mathrm{W{\cdot}(m{\cdot}K)^{-1}}$
Thermal conductivity, k_s	$0.62\,\mathrm{W{\cdot}(m{\cdot}K)^{-1}}$
Melting temperature, T_m	273 K
Numerical Methods	
Multiphase model	VOF with HRIC convection
Flow model	Laminar
Spatial discretization	2nd order (energy), 1st order (HRIC)
Boundary Conditions	
Left wall (minX)	Fixed temperature 300.0 K
Right wall (maxX)	Adiabatic
Top/Bottom walls	Adiabatic
All walls	No-slip

Table 1: CFD simulation parameters for Stefan melting-solidification problem

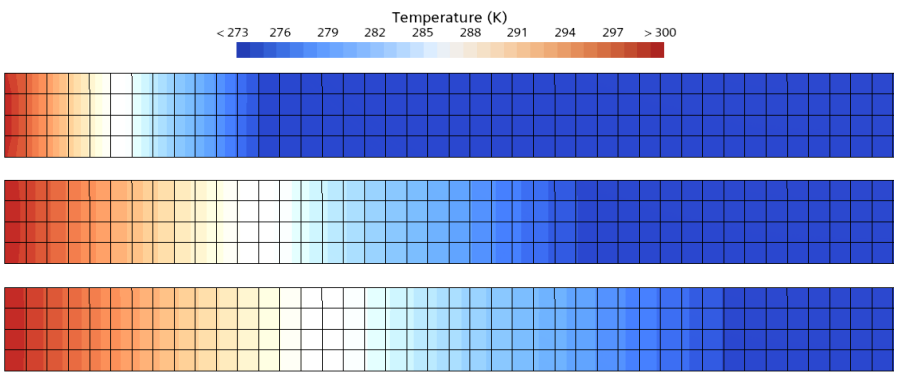


Figure 1: Evolution of the phase change interface

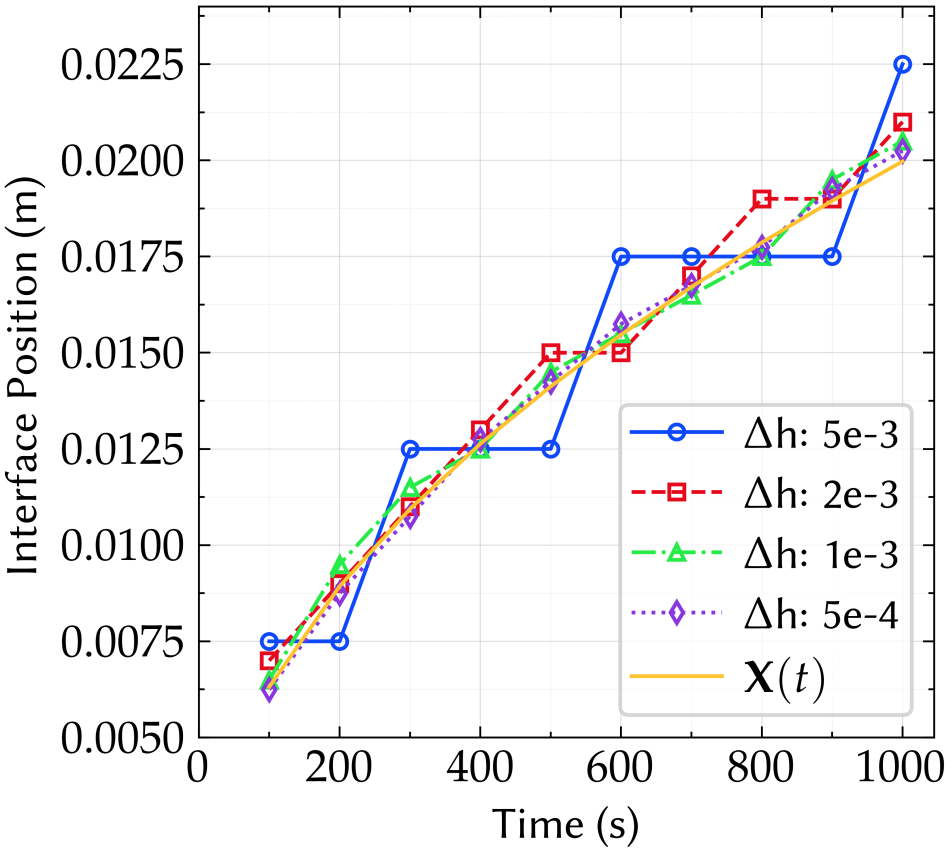
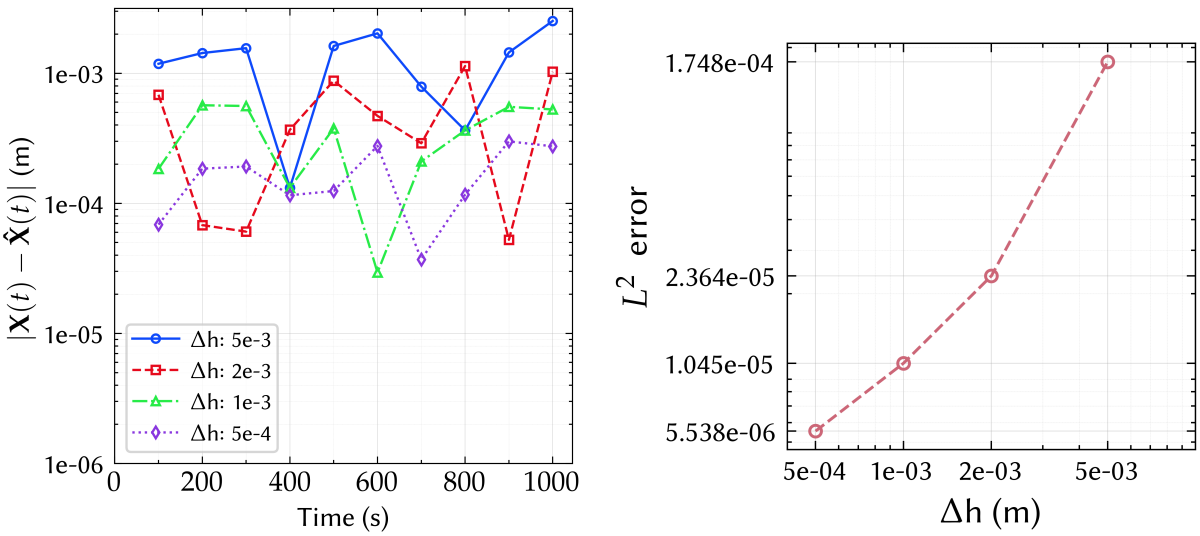


Figure 2: Location of the phase change interface



(a) Absolute error in the location of the interface

(b) Temperature profile convergence analysis

Figure 3: Error Analysis for the Stefan problem for different mesh sizes

Figure 3 shows the location of the interface for simulations with different cell sizes against the analytical solution. The location of the interface is inferred using the volume fraction of the liquid and solid phases present along with the temperature results. The numerical values are plotted at time intervals of 100 s and the absolute error in the location of the interface can be seen in Figure 3a. The temperature profiles obtained numerically can

also be compared against the analytical temperature profile in order to calculate the L_2 and L_∞ errors, where $L_2 = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(T_i - T_{analytical,i})^2}$ and $L_\infty = \max_i |T_i - T_{analytical,i}|$ and evaluate the convergence rate.

The errors in the temperature profiles for the various mesh sizes are plotted in Figure 3b. It can be seen that the simulation results converge to the analytical solution with decreasing mesh size.

2.2.2 Rayleigh-Taylor Instability

The Rayleigh-Taylor (RT) instability occurs when a denser fluid is accelerated into a lighter one, leading to interfacial perturbations and complex vortex formation [23]. This phenomenon, known as the Rayleigh-Taylor (RT) instability, is fundamental to many natural and engineering systems. Although the ideal, undisturbed RT instability is rarely encountered in practice, its essential features arise across a broad range of spatial and temporal scales—where the basic mechanism can be influenced by viscosity, surface tension, compressibility, external accelerations, or stratification[24]

Perturbation growth is driven by oppositely directed gradients of pressure and density at the interface. The density contrast between the two fluids is represented by the non-dimensional *Atwood number*, $At = (\rho_H - \rho_L)/(\rho_H + \rho_L)$, where ρ_H and ρ_L denote the densities of the upper (heavy) and lower (light) fluids respectively. The Atwood number quantifies the strength of the instability, with larger values corresponding to greater density contrasts.

The nonlinear development of the RT instability is central to a wide class of interfacial flows where one fluid penetrates or displaces another [25]. Accurately simulating RT instability serves as a classical benchmark for two-phase flow solvers, requiring detailed computation of interface dynamics and rigorous implementation of interphase interaction terms as discussed in [chapter stuf], and is widely used to validate numerical methods for multiphase flows [26].

Considering a region, $\Omega \in \mathbb{R}^2$, where two immiscible, viscous, and incompressible fluids occupy the regions Ω_H and Ω_L, such that $\Omega = \Omega_H \cup \Omega_L$. The regions $\Omega_{H,L}$ are defined using a scalar height function $h(x,t)$ that graphs the vertical position of the interface at each point along the x-axis.

$$\Omega_i(t) = \{(x,y) \in \mathbb{R} \times \mathbb{R} : (-1)^i(y - h(x,t)) > 0, t \geq 0\},\ i = H, L$$

The interface, $X(t)$, separating the two immiscible fluids is given as

$$X(t) = \{(x,y) \in \mathbb{R} \times \mathbb{R} : y = h(x,t),\ t \geq 0\}.$$

With the initial position of the interface denoted by $X(t = t_0) = X_0$, and the initial velocity field by $\underline{\mathbf{v}}(t = t_0) = \underline{\mathbf{v}}_0$, the flow is governed by the incompressible Navier-Stokes equations, surface tension σ, and acceleration $\underline{\mathbf{g}}$. The jump conditions at the interface preserve the velocity field and relate the normal stresses at the interface to the mean curvature.

If density ρ can be assumed to be $\rho_{H,L}$ or the mixture density in cases of the two-fluid and VOF methods respectively. The same can be extended to the velocity, $\underline{\mathbf{v}}$ and the dynamic viscosity μ.

The system of equations governing this two-phase flow system can be expressed as:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \rho \underline{\mathbf{v}} = 0 \qquad \text{in } \Omega_i(t) \tag{2.4}$$

$$\frac{\partial (\rho \underline{\mathbf{v}})}{\partial t} + \nabla(\rho \underline{\mathbf{v}} \otimes \underline{\mathbf{v}}) = -\nabla p + \mu \nabla^2 \underline{\mathbf{v}} + \rho \underline{\mathbf{g}} \quad \text{in } \Omega_i(t) \tag{2.5}$$

$$\nabla \cdot \underline{\mathbf{v}} = 0 \tag{2.6}$$

$$[[\underline{\mathbf{v}}]] = 0 \tag{2.7}$$

$$[[\left(\mu(\nabla\underline{\mathbf{v}} + (\nabla\underline{\mathbf{v}})^T) - (p + \rho\underline{\mathbf{g}})\underline{\underline{\mathbf{I}}}\right)\underline{\mathbf{n}}]] = \sigma\kappa\underline{\mathbf{n}} + [[\rho]]\underline{\mathbf{g}} \tag{2.8}$$

$$\underline{\mathbf{v}}(\underline{\mathbf{x}}, t_0) = \underline{\mathbf{v}}_0 \tag{2.9}$$

$$X(t_0) = X_0 \tag{2.10}$$

where $\underline{\mathbf{n}}$ is the normal vector to the interface, p is the pressure field, $\underline{\underline{\mathbf{I}}}$ is the second-order identity tensor, and κ in this case denotes the mean curvature of the interface.

In order to simulate a single mode perturbance to the interface, the initial position of the interface is defined using the amplitude, $A = 0.1L_x$, and wavelength, $\lambda = L_x$ as follows

$$X_0 = 2L_x + A\cos\left(\frac{2\pi x}{L_x}\right)$$

The dimension L_x is the characteristic length which is chosen to be the width of the simulated domain.

The numerical simulations assume turbulent and unsteady flow. The computational domain is a two-dimensional rectangle, $[0, L_x] \times [0, 4L_x]$ that is discretised using uniform orthogonal quadrilateral cells. Convective and diffusive terms are discretized by the second-order upwind scheme; cell-center gradients are interpolated to face centers using the least-squares method. Interface capturing is performed with the volume-of-fluid (VOF) method using the first-order HRIC scheme. The transient term is approximated using the first-order implicit Euler method. Turbulence effects are modeled using the Shear Stress Transport (SST) k-ω [better word?] closure, and the surface tension force is treated semi-implicitly. Gravitational acceleration acts downward, $g = 1.0\,\mathrm{m/s^2}$. Periodic boundary conditions are imposed in the horizontal direction and slip walls at the top and bottom.

The DNS study undertaken in [27] is taken to be the canonical benchmark and the physical parameters are chosen to maintain dynamic similarity.

According to [18], a reasonable solution to the RT instability with $At \approx 0.75$ can be obtained with a 128×512 mesh. With similar mesh resolutions being used in [27, 28] to study similar RT instabilities, the domain is discretised using a uniform orthogonal quadrilateral mesh with size $\Delta_h = 4.5 \times 10^{-3}$ m. Additional mesh sizes are evaluated in order to study mesh convergence. Time is non-dimensionalised using the Atwood number and the timestep for the simulations, similar to [29, 27], is selected as $\Delta t = 2.5 \times 10^{-4}/\sqrt{At}$ s.

The physical parameters and the simulation setup, along with the various cases studied are summarised in Tables 2 and 3.

Spatial and Temporal Discretisation	
Domain	2D, $[0, L_x] \times [0, 4L_x]$, where $L_x = 1\,\mathrm{m}$
Time discretization	Implicit unsteady, 1st order
Time step, Δt	$3.5 \times 10^{-3}\,\mathrm{s}$
Physical Properties	
Amplitude of perturbation, A	0.1
Wavelength of perturbation, λ	$L_x = 1.0\,\mathrm{m}$
Surface tension, σ	$1.0 \times 10^{-5}\,\mathrm{N{\cdot}m^{-1}}$
Gravity, $\underline{\mathbf{g}}$	$[0, -1.0]\,\mathrm{m{\cdot}s^{-2}}$
Numerical Methods	
Multiphase model	VOF with HRIC convection
Flow model	Turbulent, SST k–ω
Spatial discretization	2nd order (flow), 1st order (HRIC)
Gradient reconstruction	Least-squares
Boundary Conditions	
Top/Bottom walls	Wall with slip, contact angle
Left/Right boundaries	Symmetry plane

Table 2: CFD simulation parameters for Rayleigh-Taylor instability study

	At = 0.2			At = 0.5			High At		
Parameter	Re300	Re1000	Re3000	Re300	Re1000	Re3000	At30	At1000	At1500
ρ_H [kg·m^{-3}]	1.5	1.5	1.5	3.0	3.0	3.0	3.0	1.0	1.5
ρ_L [kg·m^{-3}]	1.0	1.0	1.0	1.0	1.0	1.0	0.1	0.001	0.001
$\mu_{H,L}$ [Pa·s]	0.005	0.0015	0.0005	0.01	0.003	0.001	0.001	0.001	0.001
Mesh size Δh [m]	0.0045	0.0045	0.0045	(0.01, 0.007, 0.0045)	0.0045	(0.01, 0.007, 0.0045)	0.0045	0.0045	0.0045

Table 3: Fluid properties and mesh resolutions for all simulation cases

Figure 4 depicts the evolution of the RT instability for Reynolds numbers 300, 1000, and 3000. The initial evolution of the interface is through exponential growth of the infinitesimal disturbance of the interface [30]. The growth follows linear stability theory which transitions into increasing non-linear behaviour as the amplitude grows to orders of 0.1λ [24]. This non-linear evolution is marked by the development of the distinct "bubbles" of lighter fluid rising as the "spikes" of the heavier fluid fall down. The evolving interface experiences increasing vorticity and the increasing velocity shear results in the development of Kelvin-Helmholtz type instabilities [27, 25] leading to the roll-up of the spikes as the amplitude grows to the order of $0.5 - 1\lambda$. For the case of $Re = 3000$, these instabilities are noticeable much earlier all along the interface, leading to a secondary perturbation along the boundary of the domain. Further growth leads to secondary vortices along the spike ends which leads to chaotic mixing and complex flow features. [24].

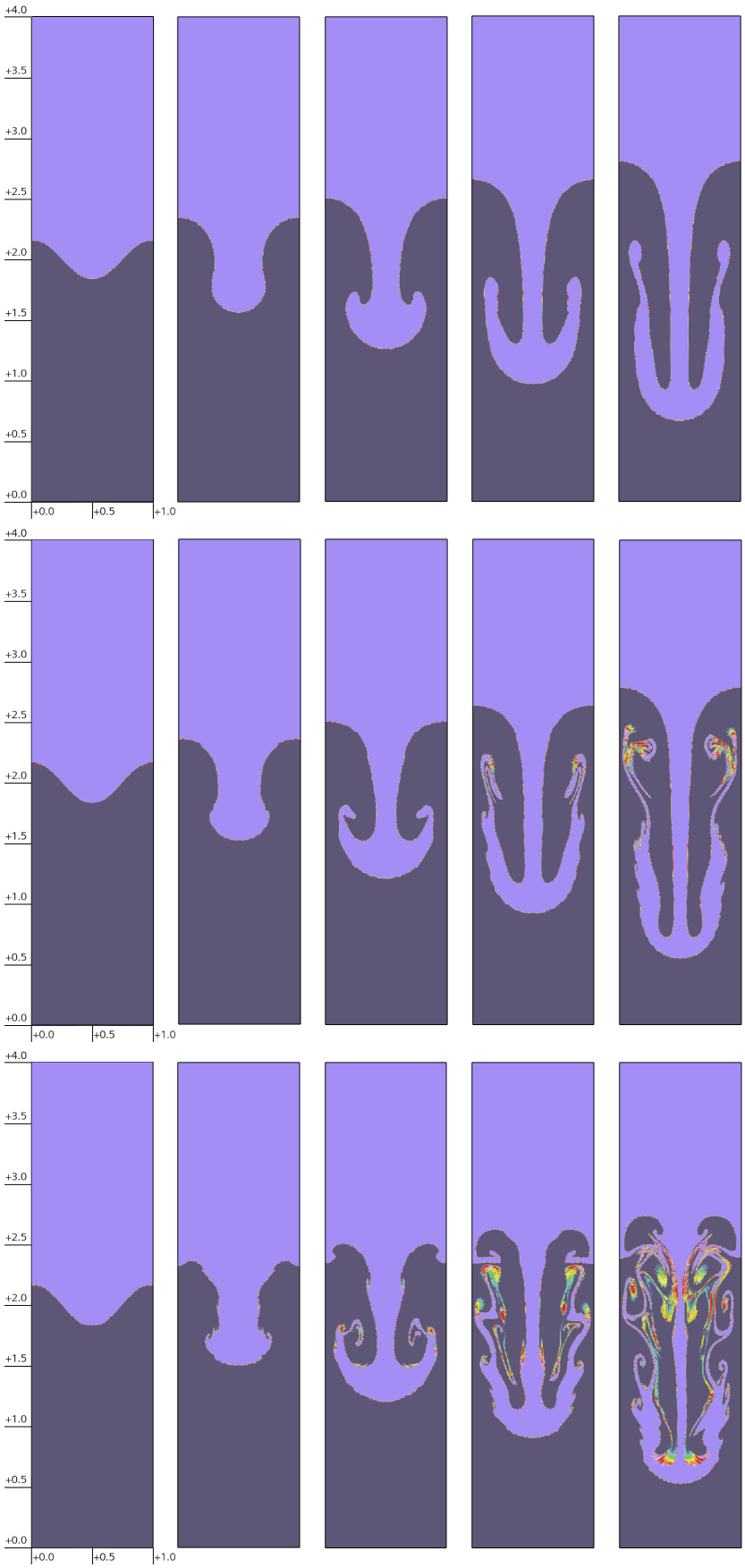


Figure 4: Evolution of Rayleigh-Taylor instability for Atwood number At = 0.5 at different Reynolds numbers.

- **Effect of Reynolds and Atwood Number:** Increasing the Reynolds number has a relatively small effect on the penetration lengths of the spikes and bubbles - with $Re = 300$ showing the least penetration depths. However, it does have a significant effect on the topology of the interface, with increasingly chaotic vortices visible at the spike rollup and the ensuing turbulent mixing. At $Re = 300$, it is seen in Figure 4, that the flow remains relatively laminar with smooth interfaces, whereas $Re = 3000$ shows significant turbulent mixing and vortical complex structures. High Atwood numbers promote non-linearity

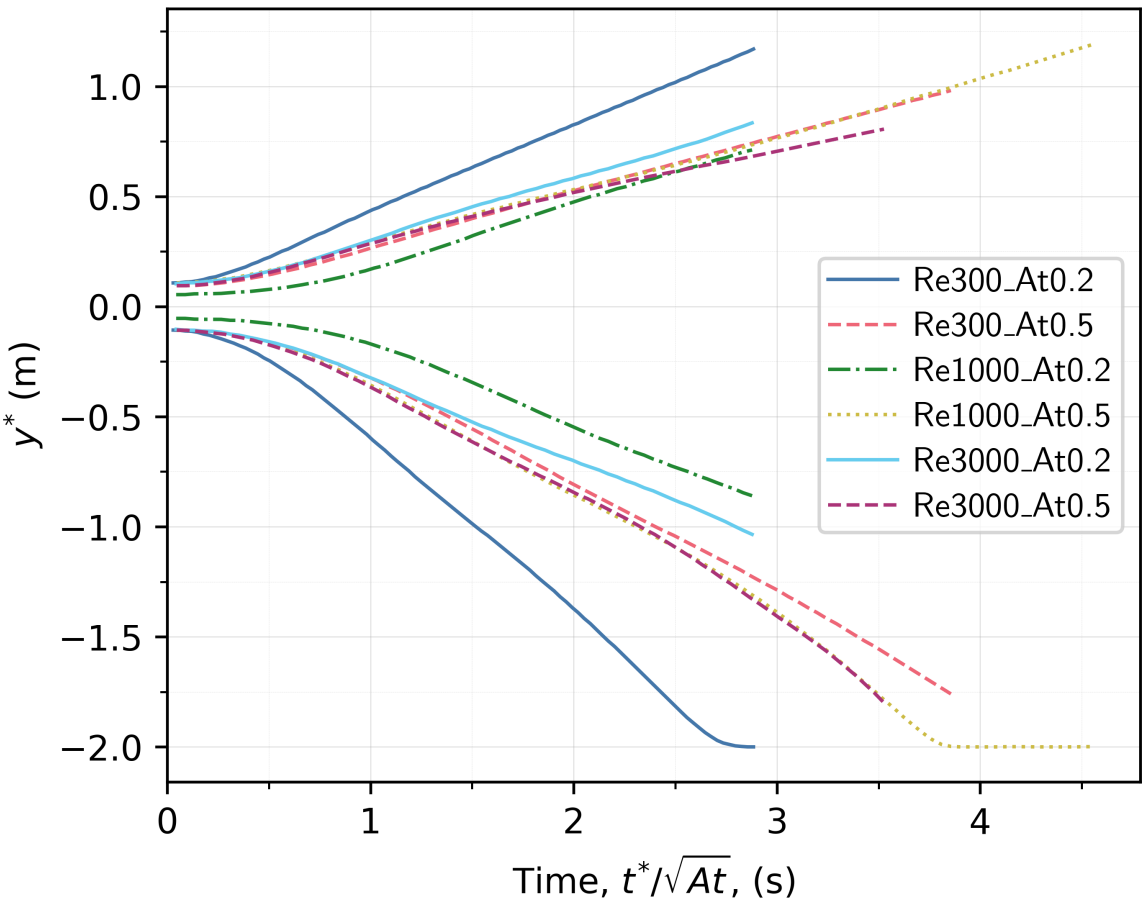
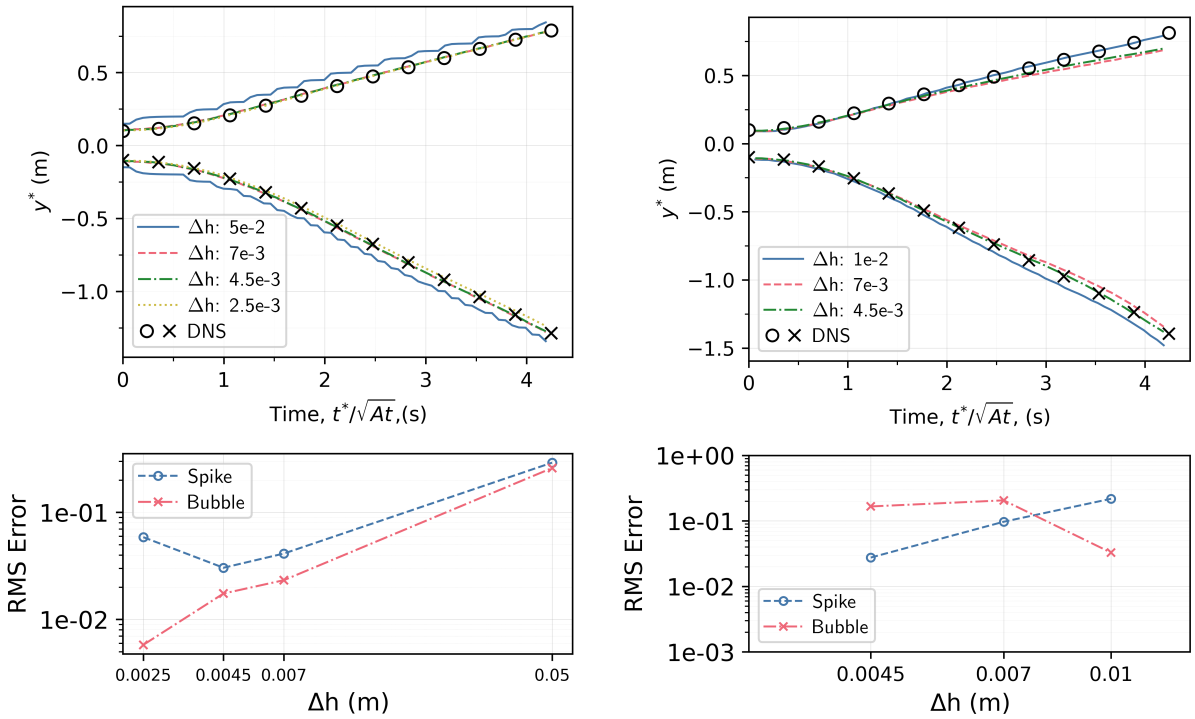


Figure 5: Evolution of the RT instability with varying Reynolds and Atwood numbers

and increased turbulent mixing, and affect the initial interface growth. The effect and prominence of the counter-rotating vortices at the tip of the spike decreases and reduces the chaotic flow features at the tips of the secondary vortices. The spike also falls much quicker. This is due to the density-driven acceleration outweighing the viscous effects as is seen in Figure 8 and in the very high density ratio simulations of the RT instability in [28].

- **Grid Convergence:** In order to evaluate the quality of the mesh, the maximum vertical distance of the spike and bubble tip as measured from the undisturbed initial interface, $y^* = |y_{s,b} - y_0|$, are evaluated for Reynolds numbers, $Re = 300, 3000$, and Atwood number, $At = 0.5$. Locations of the spike and bubble tips are recorded at every time-step and extrapolated in order to compare to the DNS data available from [27].

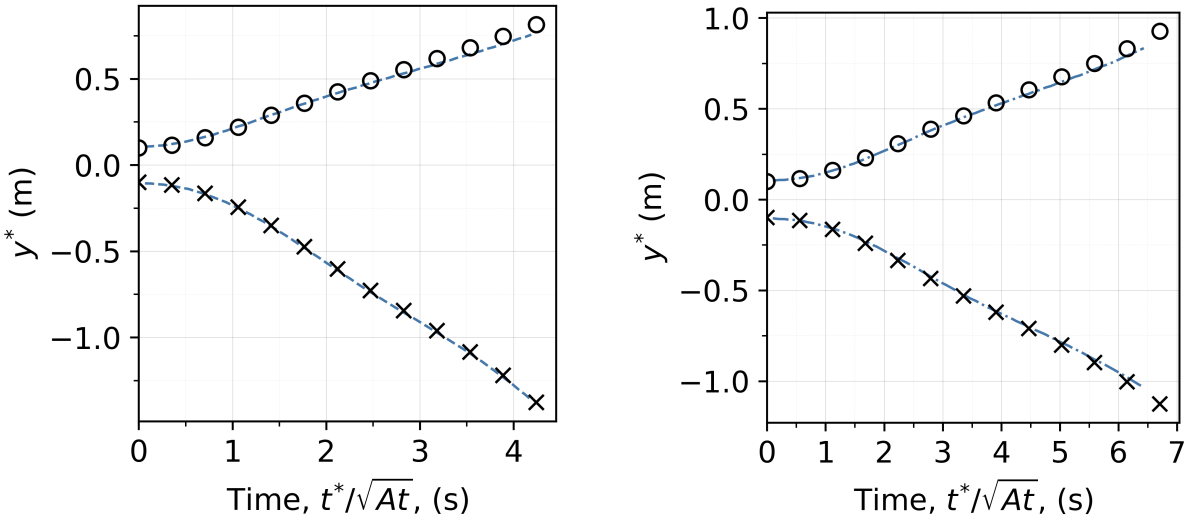
Figure 6 shows the comparison the results of the simulations with the DNS location of the bubble and spike tips along with the root mean square defined as $err = \sqrt{\sum_{t_0}^{t_f}(y_b - y_{dns})^2}$. The RMS errors and the maximum absolute errors are further summarized in table [table].



(a) $Re = 300$

(b) $Re = 3000$

Figure 6: Grid convergence analysis for Rayleigh-Taylor instability



(a) $Re = 1000$, $At = 0.5$

(b) $Re = 3000$, $At = 0.2$

Figure 7: Comparison with DNS results for different Reynolds numbers and Atwood numbers

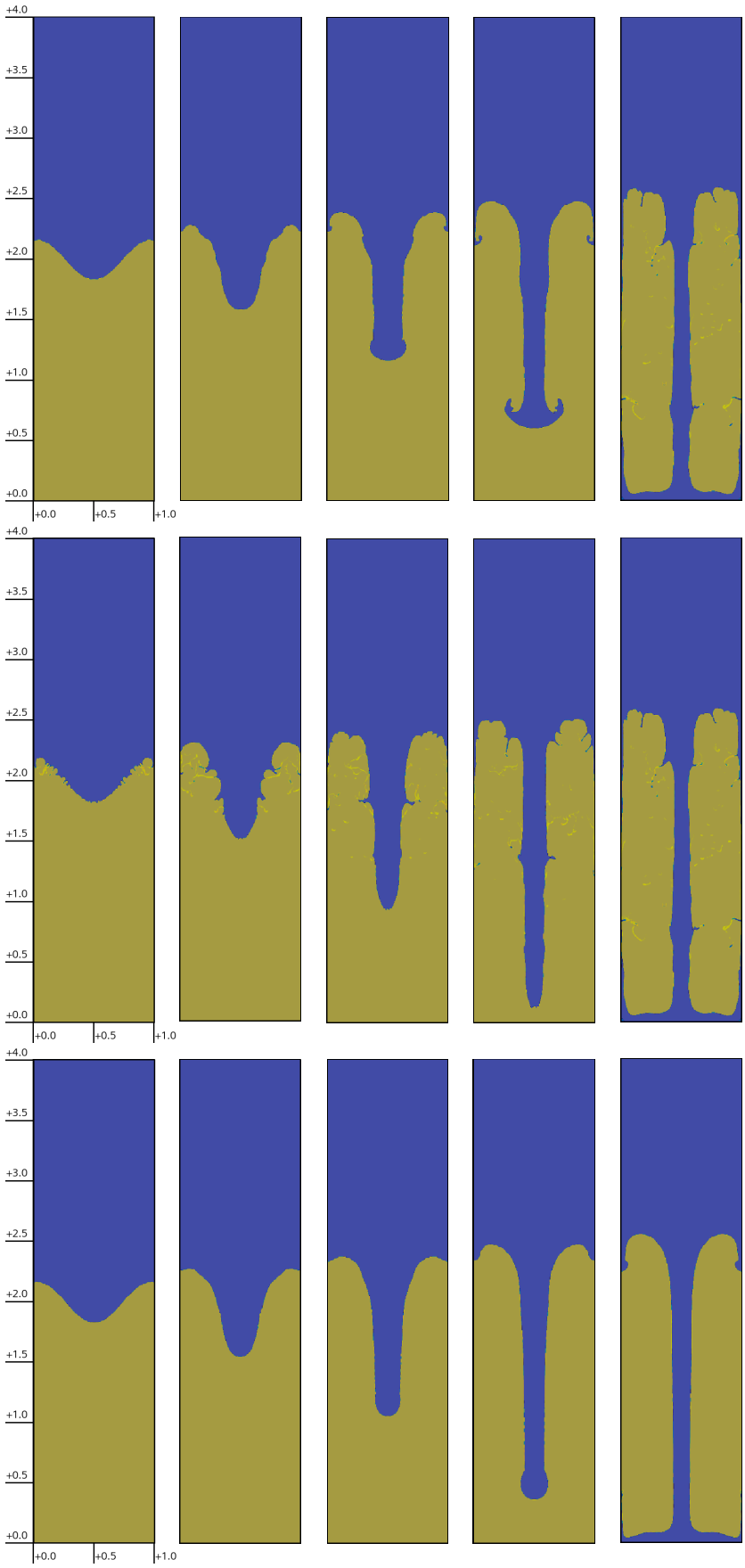


Figure 8: Evolution of Rayleigh-Taylor instability for high Atwood numbers At = 30, 1000, and 1500.

Figure 7 compares the RT instabilities for $Re = 1000, 3000$ and $At = 0.5, 0.2$ respectively without the evaluation of the errors.

Although, Star-CCM+, qualitatively, accurately simulates the RT instability, the calculated RMS and absolute errors in the bubble and spike tip locations show erratic behaviour with decreasing mesh size. For $Re = 300$, the sudden increase in the RMS error at $\Delta h = 0.0025$ can be a result of [?] For higher Reynolds numbers, the bubble tip distances are consistently under-predicted for $Re = 1000, 3000$ at later simulation times. For $Re = 3000$, at later simulation times, secondary perturbations develop at the domain edges. These secondary perturbations develop into secondary spikes that interfere with the turbulent vortices at the spike tips and the main spike stem. This is, interestingly, not seen for the coarsest mesh size for $Re = 3000$. This could be developing Kelvin-Helmholz instabilities at the surface that transition into secondary RT instabilities [31]. Although Kelvin-Helmholz instabilities develop along the interface at higher Reynolds number, this secondary perturbation is first observed for $Re = 10000$ in [27].

This can be ascribed to either the lower order scheme used in the Star-CCM+ simulations or the choice of turbulence modelling using eddy viscosity models instead of completely resolving turbulence. For Atwood number that are not close to zero, the Boussinesq approximation is also invalid which is an integral assumption in RANS turbulence models. Star-CCM+, in order to accommodate the spurious currents that may be introduced through the use of eddy viscosity models, provides models that aid in the damping of momentum and/or turbulence at the interface, however, the addition of these models leads to significant deviations from the DNS results.

2.3 Liquid Infiltration of Evaporation Chamber

The evaporation chamber and the nozzle are modelled together as a 2D domain as shown in Figure 9. Line probes present along the mid-section of the evaporation chamber and at various points along the nozzle measure the flow quantities as would be measured in the actual experiment. Star-CCM+, during the creation of a 2D domain, prescribes an automatic, unchangeable depth of $1m$ to the model. It is assumed that approximately $150mL$ water is discharged into the evaporation chamber during the experiment. In order to maintain dynamic similarity, the mass flow rate is compared between the 3D experiment and the 2D model.

An adaptive trimmed cell mesh is used to mesh the computational domain.

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- [ON EVOLUTION OF TEMPERATURE FIELD]

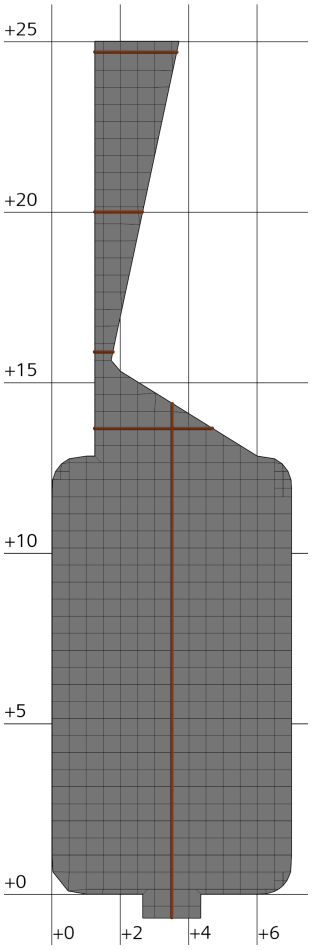

Figure 9: Computational mesh for the evaporation chamber and nozzle assembly

3 Subproblem: Evaporation

3.1 Objective

3.2 Validation Problems

3.2.1 Stefan Problem

3.3 Evaporation of Superheated Water in the Chamber

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